

MGMS IM GF USHAKOVA, Russian Federation

WMO: 200870

Lat: 79.551N

Lon: 90.572E

Elev: 27

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-36.9	-33.4	-41.1	0.5	-36.6	-37.5	0.6	-33.1	33.9	-5.7	30.8	-7.2	8.5	140	1.180

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.1	39.4	37.6	37.4	36.2	36.0	35.1	37.8	39.0	36.4	37.3	35.3	35.9	13.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
36.6	31.8	38.0	35.5	30.4	36.6	34.5	29.2	35.5	14.2	39.2	13.7	37.4	13.2	36.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.8	28.3	25.4	-38.3	43.6	5.7	3.8	-42.3	46.3	-45.6	48.6	-48.8	50.7	-52.9	53.5		
(5)			-38.3	41.1	5.5	3.7	-42.2	43.7	-45.4	45.9	-48.4	47.9	-52.4	50.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-13.2	-14.0	-12.1	-1.0	15.4	29.8	33.2	32.6	28.4	17.6	1.7	-8.1
	DBStd	20.45	12.62	12.63	11.71	10.29	7.84	3.61	2.46	2.40	4.62	9.98	11.39	12.21
	HDD50	14849	1959	1793	1926	1530	1073	606	520	540	647	1004	1449	1802
	HDD65	20324	2424	2213	2391	1980	1538	1056	985	1005	1097	1469	1899	2267
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	12.8	13.3	12.9	12.9	12.5	11.2	11.5	11.2	11.8	13.7	15.4	13.8	14.0
Precipitation	PrecAvg	6.9	0.4	0.4	0.4	0.3	0.4	0.6	1.1	1.1	0.9	0.6	0.4	0.4
	PrecMax	15.2	2.0	0.9	1.0	2.0	1.2	2.2	5.8	4.8	1.8	0.9	0.7	1.1
	PrecMin	3.3	0.0	0.1	0.0	0.0	0.1	0.1	0.0	0.1	0.1	0.1	0.1	0.1
	PrecStd	2.4	0.4	0.2	0.2	0.4	0.3	0.5	1.1	0.8	0.5	0.2	0.2	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.7	25.7	26.4	30.5	32.9	37.8	44.3	40.6	37.0	33.5	30.4	27.3
		MCWB	27.2	25.0	25.7	29.6	32.0	35.5	41.2	38.8	36.6	32.4	29.4	26.6
	2%	DB	20.6	18.4	19.5	23.1	31.0	35.9	40.7	38.4	35.8	32.6	28.6	22.9
		MCWB	20.0	17.8	18.9	22.3	30.1	34.2	38.1	37.3	35.4	31.8	27.9	22.2
	5%	DB	12.8	11.5	12.6	17.6	28.9	34.8	38.6	36.8	34.6	31.3	25.4	16.8
		MCWB	12.4	11.0	12.1	17.1	27.9	33.5	36.9	35.9	34.2	30.6	24.7	16.3
	10%	DB	4.3	4.3	4.9	13.1	26.7	33.9	36.6	35.7	33.5	29.4	19.4	8.9
		MCWB	3.7	3.8	4.4	12.2	25.8	32.9	35.4	35.0	33.0	28.5	18.6	8.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.1	25.1	25.7	29.5	32.4	35.9	41.5	39.3	36.8	33.1	29.9	26.9
		MCDB	27.5	25.8	26.3	30.2	32.7	37.0	43.9	40.0	36.9	33.3	30.1	27.2
	2%	WB	20.1	17.9	19.0	22.5	30.3	34.6	38.4	37.5	35.6	32.1	27.9	22.3
		MCDB	20.5	18.5	19.5	23.1	31.0	35.5	40.1	38.3	35.8	32.5	28.8	22.8
	5%	WB	12.4	11.0	12.2	17.0	28.1	33.8	36.9	36.2	34.3	30.6	24.6	16.3
		MCDB	12.9	11.4	12.8	17.6	28.9	34.4	38.2	36.8	34.5	31.2	25.4	16.8
	10%	WB	3.9	3.8	4.5	12.4	25.8	33.1	35.5	35.1	33.2	28.7	18.8	8.4
		MCDB	4.4	4.4	4.9	13.1	26.7	33.7	36.4	35.5	33.4	29.4	19.5	8.9
Mean Daily Temperature Range	5% DB	MDBR	10.6	10.1	10.5	9.6	7.0	3.7	3.1	3.1	3.4	6.0	8.8	9.7
		MCDBR	16.7	17.2	17.0	12.1	6.4	4.1	6.4	4.6	2.9	3.8	8.7	16.6
		MCWBR	16.7	17.1	17.0	12.0	6.2	3.4	5.0	4.0	2.9	4.1	8.7	16.4
	5% WB	MCDBR	16.7	17.3	17.1	12.1	6.4	3.8	5.9	4.5	2.9	4.0	8.9	16.7
		MCWBR	16.7	17.2	17.1	12.0	6.4	3.3	4.9	4.0	2.9	4.4	8.9	16.5
Clear-Sky Solar Irradiance	taub	N/A	N/A	0.245	0.289	0.314	0.307	0.327	0.314	0.257	N/A	N/A	N/A	N/A
	taud	N/A	N/A	1.934	2.008	2.058	2.192	2.289	2.407	2.273	N/A	N/A	N/A	N/A
	Ebn at Noon	0	0	154	227	248	260	247	229	171	0	0	0	0
	Edh at Noon	0	0	16	30	35	32	28	21	12	0	0	0	0
All-Sky Solar Radiation	RadAvg	0	0	258	1099	1831	2041	1563	806	255	0	0	0	0
	RadStd	0	0	26	62	127	164	196	94	39	0	0	0	0

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+3.01	+4.46	+3.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page